

GUILHEM CARMOUZE

Robotics engineering student, 3D vision and navigation

SEEKING 6-month end-of-studies internship, from March 2027. Robotics, 3D vision, autonomous navigation, robot perception, AI.

Toulouse, France
l7guilhem@gmail.com
linkedin.com/in/guilhem-carmouze
github.com/guilhem0908
guilhem0908.github.io

EDUCATION

Diplôme d'ingénieur, Robotic and Interactive Systems (SRI) 2024 TO JUNE 2027
UPSSITECH, engineering school of the University of Toulouse. French engineering degree, Master's level. Fifth and final year in 2026 to 2027.

Preparatory cycle (CUPGE) 2022 TO 2024
UPSSITECH, University of Toulouse

RESEARCH EXPERIENCE

Research intern, Computer Vision Research Team APRIL TO AUGUST 2026
Artificial Intelligence Research Center (AIRC), AIST, Tsukuba, Japan. 29-page report in English: "Creation of a 360° Navigation Dataset Using 3D Gaussian Splatting".

- Built ArtiFixer-360, a pipeline from a plain pinhole video to a 360° video: COLMAP, a 3DGRUT Gaussian scene, a world-locked rig of 14 overlapping 110° views repaired jointly by a 14B video diffusion model, then distilled back into the scene, with reference-free quality gates. +23,602 lines and 119 tests over NVIDIA's upstream ArtiFixer; runs on 4-GPU nodes of the ABCI cluster.
- Depth-aware synchronisation lowered the cross-view depth-overlap error of the repaired views by 27% (0.0340 to 0.0247) before distillation. On a 117-frame reference run the temporal warp error (MAE, raw renders against output) fell from 0.037 to 0.020. The full 154-frame run failed my own acceptance gates and led to a geometry-first redesign.
- Earlier phases: A* on a 5 cm occupancy grid and 4096 × 2048 panoramas rendered from six pinhole views, logged with position and heading, in a DISCOVERSE and MuJoCo simulation; a ROS 2 Humble interface prepared for a visual navigation model on the Kachaka robot (on main the model is not wired in; no run on the real robot is claimed).
- Also worked on SVLR (training-free visual language robotics, CNRS-AIST JRL), a modular pipeline that turns an instruction into robot actions without training.

AI tools were used for research, code and spell-checking, as declared in the report; not for running experiments or producing results.

CODE [artifixer-360-pipeline](#) [nav_3dgs_pano](#) [KachakaNavigation](#)

PROJECTS

Usine 4.0, final-year team project **IN PROGRESS** 2026 TO 2027
Industry 4.0 smart factory (Usine 4.0), with my class at UPSSITECH.

TLSe Racing, Formula Student driverless team 2025 TO 2026

- My part: the simulation and tooling layer. A 2D Pygame simulator with a fit-to-track camera, a typed CSV cone-track loader and a configurable field-of-view sensor model that selects the visible cones.
- Also worked on cone-detection models (PyTorch) and on image-processing and control modules (ROS, Python, C++). The planners are my teammates' work.

CODE [TLSe_Racing_Driverless](#) [PathPlanning](#)

Projet Fil Rouge, a real mobile robot 2024 TO 2025
Team of six: Arduino motor control, Raspberry Pi camera, LiDAR mapping with ICP, voice commands, ball tracking.

- My part: the web interface, a single-page app that drives the robot over the Web Bluetooth API; the Raspberry Pi MJPEG camera stream; the algorithm that turns the ball's image coordinates into drive commands to keep it centred; the timing of split voice commands.
- The semester before, with a classmate: a colour-ball detector in pure C11, without OpenCV.

CODE [PFR2\(team repository\)](#) [PFR](#)

SKILLS

LANGUAGES
Python, C, C++, Java (coursework), JavaScript and TypeScript

LIBRARIES
PyTorch, NumPy, SciPy, OpenCV

3D VISION
3D Gaussian Splatting (3DGRUT and 3DGUT, Splatfacto, DISCOVERSE), COLMAP, video diffusion pipelines (ArtiFixer), panoramic and equirectangular geometry

ROBOTICS
ROS 2 Humble, MuJoCo, path planning (A*, occupancy grids), Kachaka API (gRPC), robot kinematics coursework (DH model and Jacobians of a 6-DOF UR3)

TOOLS
Docker, Singularity, PBS on an HPC cluster (ABCI), Git, Linux, pytest

SIDE PROJECTS

Personal projects built in October 2026 with AI assistance; every number is reproduced by a script in the repository.

erpklt
NumPy toolkit measuring what stitching views into a 360° panorama costs.

microslat
3D Gaussian Splatting in NumPy and PyTorch, one test per equation.

gaussian-projection-bench
EWA against unscented projection error, pinhole to equirectangular.

amr-traffic-lab
Robot traffic in an Industry 4.0 smart-factory (Usine 4.0) aisle.

LANGUAGES

French native
English professional; 4-month research internship in Japan, in English
Spanish basic

OTHER EXPERIENCE

Administrative assistant
2022 TO 2024
ALTINET, Tarbes